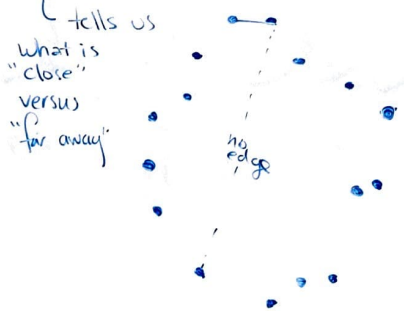
Čech Complex

P = (point) set $\subseteq (M, d)$ ← metric space

$d : M \times M \to \mathbb{R}$ distance fcn

$t \in [0, \infty)$ a threshold

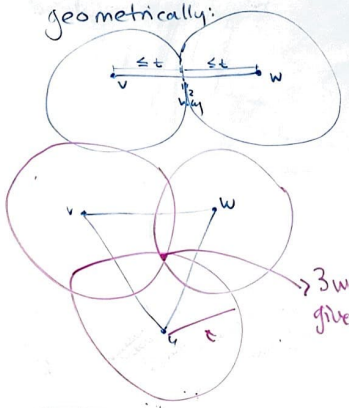
↑ tells us what is "close" versus "far away"



V = 1-1 corresp w/ elts of P

$E_t = \{(v, w) \in V \times V \mid v \neq w,\ d(v, w) \leq 2t\}$

geometrically:



→ 3 way intersection gives us 2-cells

*Both Čech and VR complexes are A.S.C. But not G.S.C. [Why?]

K-dim cells

$$K_{k,t} := \left\{ W \subseteq V \;\middle|\; |W| = k+1,\ \bigcap_{w_i \in W} B(w_i, t) \neq \emptyset \right\}$$

Q Why is this a simplicial complex?

the ball
centered at
wrt the
$B(w_i, t) :=$

olexes

$\{W \subseteq V \mid |W| = k+1, \bigcap_{w_i \in W} B(w_i, t) \neq \emptyset\}$

the ball of rad t centered at w_i, wrt the distance d

$B(w_i, t) := \{x \in M \mid d(w_i, x) \leq t\}$

y is this a simplicial
nplex?

Vietoris-Rips Cplx ← can have (P, d) or $P \subset (M, d)$

(approximation of Čech cplx)

V, E_t the same

$K_{k+1,t} = \{W \subseteq V \mid |W| = k+1, \text{diam}(W) \leq 2t\}$

$\text{diam}(W) := \max_{a,b \in W} d(a,b)$

Q Relationship between Čech + Rips?

$\check{C}(P, t) \subseteq VR(P, t) \subseteq \check{C}(P, \sqrt{2}t)$

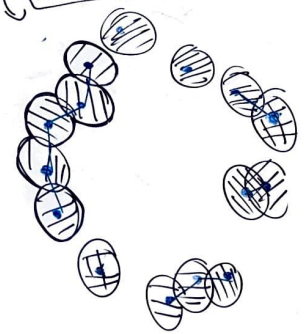
↑ Sec EH

i.e. the smallest Cplx built on V, E_t

P = pt set $\subseteq \mathbb{R}^d$



$\bigcup_{p \in P} \bar{B}(p,t)$ is a topo space!



def^n:

Nerve

$\mathcal{S}$ = a set of sets

The nerve of $\mathcal{S}$ is

$N(\mathcal{S}) = \{\sigma \subseteq \mathcal{S} \mid \bigcap_{X_i \in \sigma} X_i \neq \emptyset\}$

(σ: a set of sets)

The Čech complex is a nerve!

$\check{C}(P,r) = N\left(\{\bar{B}(p_i,t) \mid p_i \in P\}\right)$

Nerve Theorem: [DW]

If $\mathcal{S}$ is a finite set of sets in a topo space s.t.

$\forall \sigma \subseteq \mathcal{S}$, $\cap\sigma := \bigcap_{X_i \in \sigma} X_i$ (a set of sets → a set)

is either empty or compact + connected, then

$N(\sigma) \boxed{\simeq} \cup\mathcal{S} := \bigcup_{X_i \in \mathcal{S}} X_i$

$\simeq$: homotopy equiv.

Q: Find a different version of the Nerve Thm!

EH: $\mathcal{S}$ = set of convex sets

X, Y topo spaces

$g, f : X \longrightarrow Y$

f, g are homotopy equiv iff $\exists$ cont. $H : X \times [0,1] \to Y$

$(\cdot, 0) \longmapsto f(\cdot)$

$(\cdot, 1) \longmapsto g(\cdot)$

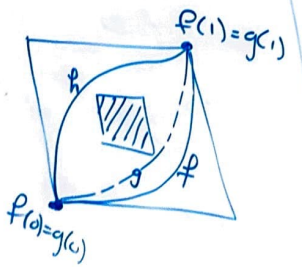
Write: $f \simeq g$.

Think of $[0,1]$ as "time" morphing one {map / path} into the other continuously.

examples

$X = [0,1] =: I$

we have paths! btwn fixed verts



X, Y topo spaces

$X \simeq Y$ iff $\exists$ cont. maps $X \underset{g}{\overset{f}{\rightleftarrows}} Y$ such that $f \circ g \simeq id_Y$ and $g \circ f \simeq id_X$

Nerve The